

AI Assisted Verification and manual Verification

Epic: AI Assisted Verification

Feature: AI Assisted Verification Hub

Product Vision

To build an AI-assisted document verification platform that automates identity, regulatory, tax, financial and internal banking verifications, enabling Credit Officers to make faster and more informed lending decisions while reducing manual effort, operational risk, and loan processing turnaround time.

Business Purpose

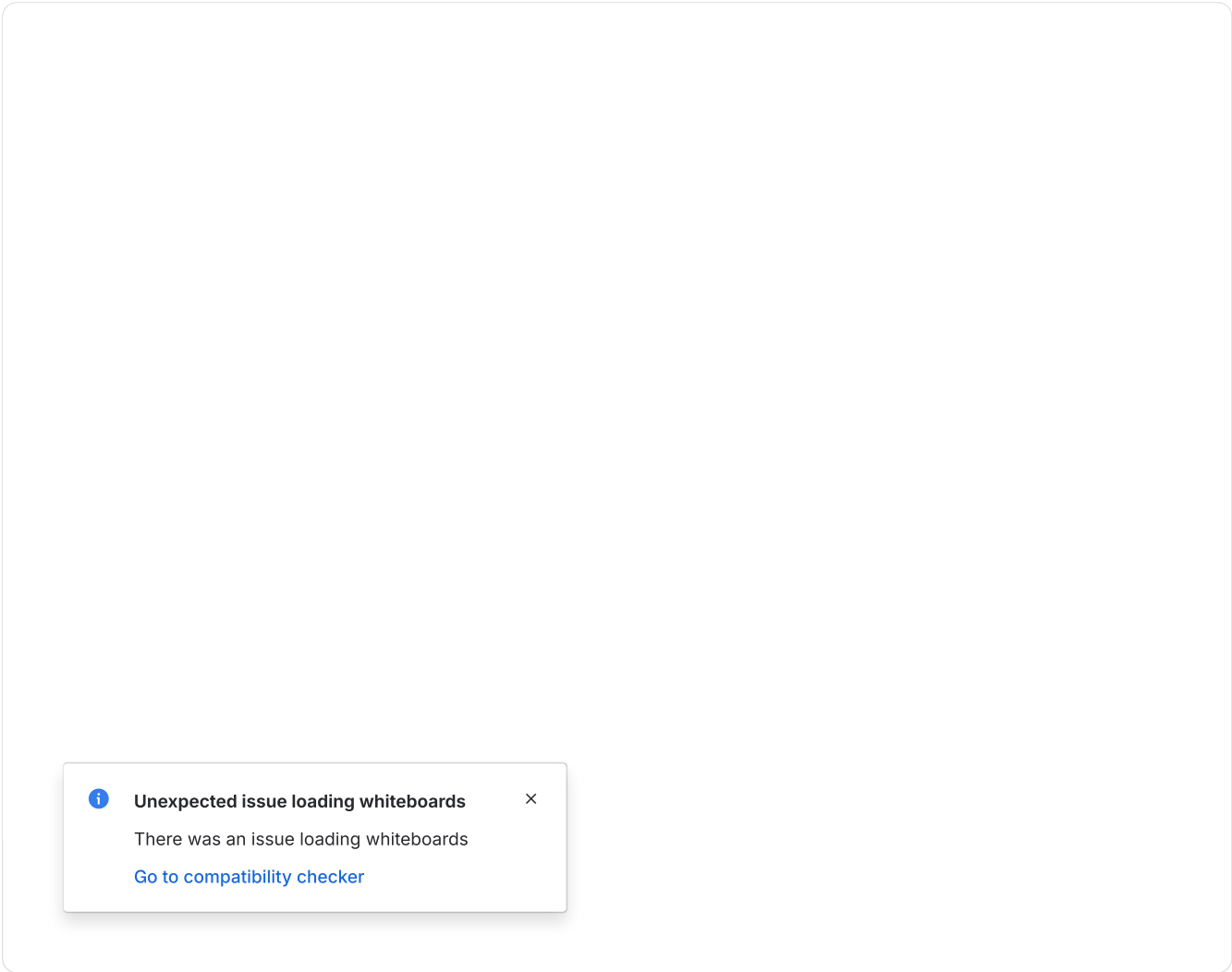
The AI Assisted Verification Hub provides a centralized verification dashboard that validates customer documents before credit assessment begins.

Instead of visiting multiple external websites and internal systems, the Credit Officer can review AI verification results from a single screen and focus only on verification exceptions requiring manual intervention.

The objective is to:

- Reduce document verification turnaround time.
- Improve operational efficiency.
- Reduce repetitive manual effort.
- Improve document authenticity validation.
- Enable Credit Officers to focus on credit assessment instead of administrative verification activities.
- Improve customer experience through faster loan processing

High-Level Business Process Diagram



ACTOR

Primary Actor

AI Verification Engine

Responsible for performing automated verification of uploaded customer documents, generating verification results, confidence scores, exception alerts and verification summaries.

Secondary Actor

Credit Officer

Responsible for reviewing AI verification results, manually validating failed or pending verifications, uploading missing documents, performing field verification and making the final credit recommendation.

Success Metrics (KPIs)

KPI	Target
Average Document Verification Time	Reduce from 10 minutes to 2–3 minutes
Loan Processing Turnaround Time (TAT)	Reduce by 40–60%
Manual Verification Effort	Reduce by 70%
Credit Officer Productivity	Increase number of applications processed per day
First-Time Verification Success	≥95%
Verification Audit Trail	100% Available
Manual Verification Exceptions	Only for failed or pending verification cases

Scope

In Scope

- PAN Verification
 - Aadhaar Verification
 - GST Registration Verification
 - GST Return Verification
 - Income Tax Return Verification
 - Audited Financial Statement Verification
 - Internal Banking Validation
 - Account Documentation Validation
 - AI Verification Summary
 - AI Confidence Score
 - AI Exception Management
 - Missing Document Detection
 - Manual Document Upload by Credit Officer
 - Manual Verification Queue
 - Audit Trail
-

Out of Scope

- Live NSDL Integration
 - Live UIDAI Integration
 - Live GSTN Integration
 - Live Income Tax Integration
 - Live CIBIL Integration
 - Live CRIF Integration
 - Loan Approval Decision by AI
 - Real-time Fraud Detection
 - AI Model Training using Customer Data
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Global Business Rules

BR-01 AI Confidence Score

The AI Verification Engine shall generate a confidence score for every verification performed.

The confidence score shall be displayed to the Credit Officer.

Confidence Score	Status	Credit Officer Action
95–100%	High Confidence	Review only
80–94%	Medium Confidence	Credit Officer may perform additional validation
Below 80%	Low Confidence	Manual verification mandatory

BR-02 AI Decision Support

The AI Verification Engine shall only perform document verification and generate recommendations.

The AI shall NOT:

- Approve Loan
- Reject Loan
- Recommend Loan Amount
- Override Credit Officer Decision

Final lending decision shall always remain with the Credit Officer.

BR-03 AI Independence

The AI Verification Engine shall process every loan application independently.

The AI shall NOT:

- Learn from previous customer data.
- Learn from previous loan applications.
- Learn from previous Credit Officer decisions.
- Modify future verification results using historical customer information.

Every verification request shall be processed independently.

BR-04 Explainable AI

For every verification, AI shall display:

- Verification Status
 - Confidence Score
 - Verification Source
 - Verification Timestamp
 - Failure Reason (if applicable)
 - Manual Action Required
-

BR-05 Manual Override

The Credit Officer shall be able to:

- Override AI verification results
- Upload missing documents
- Record remarks
- Complete manual verification
- Submit final recommendation

All overrides shall be recorded in the Audit Trail.

BR-06 Failed Verification

Whenever AI verification fails:

- Display Red Alert
 - Display Failure Reason
 - Add to Manual Verification Queue
 - Notify Credit Officer
 - Prevent workflow progression until resolved
-

BR-07 Missing Documents

If mandatory documents are missing:

- AI continues verifying available documents.
 - Missing documents shall be marked **Not Uploaded**.
 - Credit Officer shall be allowed to upload missing documents.
 - AI may re-run verification after upload (Demo).
-

User Story 1 – PAN Verification

As an AI Verification Engine,

I want to verify the uploaded PAN against the NSDL verification service (Demo)

So that

I can validate the customer's PAN authenticity and update the verification status before credit assessment.

Acceptance Criteria

Given

A PAN document has been uploaded.

When

AI initiates PAN verification.

Then

- PAN status becomes **Verified**
- Verification source displays **NSDL (Demo)**
- Verification timestamp is recorded
- AI confidence score is displayed
- Status is visible to the Credit Officer

- If confidence is below 80%, add to Manual Verification Queue
 - If verification fails, display Red Alert
-

User Story 2 – Aadhaar Verification

As an AI Verification Engine,

I want to verify the uploaded Aadhaar using UIDAI verification service (Demo)

So that

I can authenticate the customer's identity.

Acceptance Criteria

Given Aadhaar document has been uploaded.

When AI performs Aadhaar verification.

Then

- Aadhaar status becomes Verified
 - Name matches application
 - Verification source displays UIDAI (Demo)
 - Timestamp recorded
 - AI confidence score displayed
 - Low confidence routes to Manual Verification Queue
 - Failed verification displays Red Alert
-

User Story 3 – GST Registration Verification

As an AI Verification Engine,

I want to verify GST Registration using GST Portal (Demo)

So that

I can validate the business registration.

Acceptance Criteria

Then

- GST Registration Verified
- GST Active

- Legal Name Match
 - Verification Source GSTN (Demo)
 - Timestamp
 - Confidence Score
 - Low confidence → Manual Queue
 - Failed verification → Red Alert
-

User Story 4 – GST Returns Verification

As an AI Verification Engine,

I want to retrieve the last 12 months GST Returns.

Acceptance Criteria

Then

- Last 12 months GST Returns displayed
 - Filing Status displayed
 - Verification completed
 - Confidence Score displayed
 - Failed retrieval → Red Alert
 - Manual Verification Queue updated
-

User Story 5 – Income Tax Return Verification

As an AI Verification Engine,

I want to verify the uploaded Income Tax Return.

Acceptance Criteria

Then

- ITR Verified
 - Assessment Year displayed
 - Filing Status displayed
 - Timestamp recorded
 - Confidence Score displayed
 - Failed verification → Manual Queue
-

User Story 6 – Audited Financial Statement Verification

As an AI Verification Engine,

I want to verify uploaded audited financial statements.

Acceptance Criteria

Then

- Financial Statements Verified
 - Validation completed
 - Timestamp recorded
 - Confidence Score displayed
 - Failed verification → Red Alert
-

User Story 7 – Internal Banking Validation

As an AI Verification Engine,

I want to validate customer information using internal banking records.

Acceptance Criteria

Then

- Internal Banking Validation Verified
 - Banking relationship confirmed
 - Timestamp recorded
 - Confidence Score displayed
 - Failed validation → Manual Queue
-

User Story 8 – Account Documentation Validation

As an AI Verification Engine,

I want to verify all mandatory documents have been uploaded.

Acceptance Criteria

Then

- Uploaded documents marked Verified
- Missing documents marked Not Uploaded
- Exception list updated

- Credit Officer notified
 - Credit Officer may upload missing documents
 - AI may re-verify uploaded documents
-

User Story 9 – CIBIL Verification

As an AI Verification Engine,

I want to identify whether CIBIL verification is available.

Acceptance Criteria

Then

- Status displayed as Pending Manual Pull
 - Manual task assigned to Credit Officer
 - AI confidence displayed as Not Applicable
 - No live integration
-

User Story 10 – CRIF Verification

As an AI Verification Engine,

I want to identify whether CRIF verification is available.

Acceptance Criteria

Then

- Status displayed as Pending Manual Pull
 - Manual task assigned
 - No live integration
 - Credit Officer notified
-

User Story 11 – Missing Document Detection

As an AI Verification Engine,

I want to identify documents that were not uploaded.

Acceptance Criteria

Then

- Available documents continue verification
 - Missing documents marked Not Uploaded
 - Loan application not rejected
 - Credit Officer may upload missing documents
 - AI re-runs verification after upload (Demo)
-

User Story 12 – AI Verification Summary

As an AI Verification Engine,

I want to generate an AI verification summary.

Acceptance Criteria

Then

- Total Documents Received displayed
 - AI Verified Count displayed
 - Pending Manual Verification displayed
 - AI Confidence Score displayed
 - Fraud Risk displayed
 - AI Recommendation displayed
 - Summary available before manual verification
-

User Story 13 – AI Exception Management

As an AI Verification Engine,

I want to identify failed or inconsistent verification results and generate exception alerts

So that

the Credit Officer can manually validate the document before credit assessment.

Acceptance Criteria

Scenario 1 – Successful Verification

Given

The uploaded document is valid.

When

AI completes verification.

Then

- Verification status = Verified
 - Green Status displayed
 - Confidence Score displayed
 - Included in AI Summary
-

Scenario 2 – Failed Verification

Given

AI cannot verify the document.

When

Verification completes.

Then

- Status becomes Failed
 - Red Alert displayed
 - Failure reason displayed
 - Added to Manual Verification Queue
 - Credit Officer notified
 - Workflow paused until resolved
-

Scenario 3 – Missing Document

Given

A mandatory document has not been uploaded.

When

AI starts verification.

Then

- Status becomes Not Uploaded
- Available documents continue verification
- Missing document added to Manual Verification Queue
- Credit Officer can upload the document manually

- AI re-runs verification after upload (Demo)

AI Dashboard Status Behaviour

Status	Colour	Action
Verified	 Green	No action required
Pending Manual Pull	 Amber	Credit Officer to complete manual verification
Failed Verification	 Red	Immediate manual validation required
Not Uploaded	 Grey	Credit Officer may upload the document manually

AI Governance

- AI is a decision-support system only.
- AI shall not approve or reject loans.
- AI shall not learn from customer data.
- AI shall not use previous customer applications to influence future verification outcomes.
- Every loan application shall be processed independently.
- The Credit Officer retains full accountability for the final credit decision.
- All AI verification results, confidence scores, manual overrides, and final decisions shall be recorded in the audit trail for compliance, governance, and regulatory purposes.

Epic: Credit Officer Manual Verification

Feature: Credit Officer Verification

Product Vision

To provide Credit Officers with a centralized manual verification workspace where they can review AI verification results, validate exceptions, perform physical verification, upload missing documents, complete credit bureau verification, and record their final credit recommendation before the loan moves to the credit assessment stage.

Business Purpose

The Credit Officer Verification module enables Credit Officers to review AI-generated verification results and complete all manual verification activities that require human judgment or field investigation.

This module ensures that all verification exceptions are resolved before credit assessment, reducing operational risk while maintaining regulatory compliance and auditability.

Actors

Primary Actor

Credit Officer

User Story 1 – Review AI Verification Results

As a Credit Officer,

I want to review all AI verification results on a single dashboard

So that

I can quickly identify verified documents and focus only on documents requiring manual verification.

Acceptance Criteria

Given

AI verification has completed.

When

The Credit Officer opens the AI Assisted Verification Hub.

Then

- All verification results shall be displayed.
- Verification status shall be colour coded.
- Confidence score shall be displayed.
- Verification source shall be displayed.
- Verification timestamp shall be displayed.
- Failed verifications shall be highlighted.
- Manual Verification Queue shall be displayed.

User Story 2 – Review Failed AI Verification

As a Credit Officer,

I want to review failed AI verifications

So that

I can manually validate the documents before credit assessment.

Acceptance Criteria

Given

One or more AI verifications have failed.

When

The Credit Officer opens the Manual Verification Queue.

Then

- Failed verification shall be highlighted in Red.
- Failure reason shall be displayed.
- Credit Officer shall record verification remarks.
- Credit Officer shall update verification outcome.
- Audit trail shall be maintained.

User Story 3 – Upload Missing Documents

As a Credit Officer,

I want to upload missing customer documents

So that

the loan application can continue without asking the customer to restart the application process.

Acceptance Criteria

Given

AI identifies missing documents.

When

The Credit Officer receives the missing documents.

Then

- Credit Officer shall upload the document.
 - Upload timestamp shall be recorded.
 - Uploaded by shall be recorded.
 - AI shall re-run verification (Demo).
 - Audit trail shall be updated.
-

User Story 4 – Manual PAN Verification

As a Credit Officer,

I want to manually verify PAN when AI verification fails

So that

I can confirm the authenticity of the customer's PAN before credit assessment.

Acceptance Criteria

Given

PAN verification failed.

When

Credit Officer performs manual verification.

Then

- Manual verification status updated.
 - Remarks mandatory.
 - Verification date captured.
 - Audit trail updated.
-

User Story 5 – Manual GST Verification

As a Credit Officer,

I want to manually verify GST registration

So that

I can confirm business authenticity.

Acceptance Criteria

Then

- GST manually verified.
 - Remarks recorded.
 - Status updated.
 - Audit trail maintained.
-

User Story 6 – Pull CIBIL Report

As a Credit Officer,

I want to manually initiate the CIBIL bureau verification

So that

I can assess the applicant's credit profile before making a recommendation.

Acceptance Criteria

Given

CIBIL verification is Pending Manual Pull.

When

The Credit Officer clicks **Pull CIBIL Report**.

Then

- CIBIL report is retrieved (Demo).
 - Credit Score is displayed.
 - Active Loan details displayed.
 - DPD displayed.
 - Report timestamp recorded.
 - Remarks captured.
-

User Story 7 – Pull CRIF Report

As a Credit Officer,

I want to manually retrieve the CRIF bureau report

So that

I can review additional credit bureau information.

Acceptance Criteria

Then

- CRIF report displayed (Demo).
 - Credit Score displayed.
 - Delinquencies displayed.
 - Outstanding Exposure displayed.
 - Timestamp recorded.
 - Remarks recorded.
-

User Story 8 – Business Unit Verification

As a Credit Officer,

I want to record my business site visit

So that

I can verify that the business exists and is operational.

Acceptance Criteria

Given

Business verification is required.

When

Credit Officer visits the business premises.

Then

- Unit Visit Completed captured.
 - GPS coordinates captured.
 - Geo-tagged photographs uploaded.
 - Business operational status recorded.
 - Stock verification recorded.
 - Machinery verification recorded.
 - Neighbour enquiry completed.
 - Remarks mandatory.
-

User Story 9 – Residence Verification

As a Credit Officer,

I want to record my residence verification

So that

I can confirm the applicant's residential address.

Acceptance Criteria

Then

- House visit recorded.
 - GPS location captured.
 - Geo-tagged photographs uploaded.
 - Residence verified.
 - Remarks recorded.
 - Timestamp recorded.
-

User Story 10 – Manual Verification Override

As a Credit Officer,

I want to override AI verification results where appropriate

So that

I can complete the loan assessment based on my professional judgment and supporting evidence.

Acceptance Criteria

Given

AI verification differs from manual findings.

When

The Credit Officer performs an override.

Then

- Override reason is mandatory.
- Supporting remarks are mandatory.
- Manager approval required (if applicable).

- Audit trail updated.
 - Original AI result retained for reference.
-

User Story 11 – Credit Officer Recommendation

As a Credit Officer,

I want to submit my recommendation after completing all required verifications

So that

the application can move to the credit assessment stage.

Acceptance Criteria

Given

All mandatory verification activities are completed.

When

The Credit Officer submits the recommendation.

Then

The following options shall be available:

- Approve
- Approve with Conditions
- Send Back
- Reject

The system shall require:

- Recommendation Remarks
- Verification Date
- Credit Officer Name

The application status shall be updated.

User Story 12 – Audit Trail

As a Credit Officer,

I want every manual verification activity to be recorded

So that

all verification activities are traceable for audit, compliance, and regulatory purposes.

Acceptance Criteria

Then

The system shall record:

- Verification Activity
- Verification Date & Time
- User ID
- Officer Name
- Previous Status
- New Status
- Remarks
- Override Details (if applicable)

Audit records shall not be editable or deletable.

Business Rules

- The Credit Officer cannot bypass mandatory manual verification for failed AI checks.
- Manual verification is mandatory for all **Red Alert** items.
- Credit bureau verification (CIBIL/CRIF) must be completed before the final recommendation.
- GPS and geo-tagged photographs are mandatory for business and residence visits.
- Missing documents may be uploaded by the Credit Officer on behalf of the customer.
- Every manual action must be logged in the audit trail.
- The Credit Officer may override AI results only with mandatory remarks and, where required, managerial approval.
- The final credit recommendation remains the responsibility of the Credit Officer; AI serves only as a decision-support tool.
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